

Introduction

Filariasis. *Ślīpada* is a progressive swelling of lower limb which generally develops after the attack of fever. In this indisposition due to solid oedema of the foot and leg the patient becomes hard like stone. Sometimes the indisposition also affects hand and scrotum Elephantiasis-Morbid enlargement of the leg, swelled leg.

श्लीपद हेतु (Ślīpada Hetu)

पारियात्रभवा याश्च विन्ध्यसह्यभवाश्च याः ।

शिरोहृद्रोगकुष्ठानां ता हेतुः श्लीपदस्य च ॥

C.S.Su. 27/212

Rivers which originates from *Pariyātra* (Western *vindhya* range), *Vindhya*, *Sahya* ranges causes the diseases of head, heart, *Kuṣṭha*, & *Ślīpada*.

महेन्द्रप्रभवाः श्लीपदोदराणि, हिमवत्प्रभवा हृद्रोगश्च यथु-
शिरोरोगश्लीपदगलगण्डान् । S.S.Su. 45/21

Water of rivers arising from *Mahendra* mountains (North range of Western ghat) produces *Ślīpada* & *Udara*. Water of rivers arising from *Himvat* (*Himālaya*) mountains causes *Hṛdroga*, *Śvayathu*, *Śīroroga*, *Ślīpada* and *Galagaṇḍa*.

पुराणोदकभूयिष्ठाः सर्वतुषु च शीतलाः ।

ये देशास्तेषु जायन्ते श्लीपदानि विशेषतः ॥

S.S.Ni. 12/17

Ślīpada develops in those places which have a collection of stagnant water and cold climate in all the seasons.

श्लीपद संप्राप्ति (Samprāpti of Ślīpada)

षष्ठी रोहिणी नाम ब्रीहिप्रमाणा, ग्रन्थ्यपच्यर्बुदश्लीपद-
गलगण्डाधिष्ठाना । S.S.Śā. 4/4

Sixth layer of skin named *Rohiṇī* equal to one grain of paddy in thickness, is the seat of *Granthi*, *Apaci*, *Arbuda*, *Ślīpada* & *Galagaṇḍa*.

जङ्घासु पिण्डीप्रपदोपरिष्ठात् स्याच्छ्लीपदं मांसकफाम्बुदोषान् ।

C.S.Ci. 12/98

Ślīpada manifest by vitiating *Māmsa*, *Kapha* & *Rakta*. It is characterised by swelling in the calf region, lower extremities & swelling originates from the portion of the leg.

कुपितास्तु दोषा वातपित्तश्लेष्माणोऽधः प्रपन्ना वङ्गणोरुज्जानुज-
ङ्घास्ववतिष्ठमानाः कालान्तरेण पादमाश्रित्य शनैः शोफं जनयन्ति,
तं श्लीपदमित्याचक्षते । तत् त्रिविधं-वातपित्तकफनिमित्तमिति ॥

S.S.Ni. 12/10

प्रस्थिता वङ्गणोर्वादिमधः कायं कफोत्त्रणाः ॥

दोषा मांसास्त्रगाः पादौ कालेनाश्रित्य कुर्वन्ते ।

शनैः शनैर्घनं शोफं श्लीपदं तत्प्रचक्षते ॥

A.H.Utt. 29/18-19

Excessively excacerbated *Vāta*, *Pitta*, & *Kapha* move downwards and get localised in the groins, thighs, foreleg, & calves, gradually reach the foot in due course of time & gives rise to swelling is known as *Ślīpada*.

Nidāna Sevana (Consumption of Etiological factors)



Aggravation of *Kapha pradhan Tridoṣa*



Vitiates *Māmsa* & *Rakta*



Move downwards



Get localised in the groins, thighs, foreleg, calves and gradually reach the foot



Swelling in the lower limbs



Ślīpada

भेद (Bheda)

Ślīpāda is classified into three types namely *Vātaja*, *Pittaja*, and *Kaphaja*.

त्रीण्यप्येतानि जानीयाच्छलीपदानि कफोच्छ्रयात् ।
गुरुत्वं च महत्त्वं च यस्मान्नास्ति विना कफात् ॥

S.S.Ni. 12/16

All varieties of *Ślīpada* are produced by *Kapha* because without *Kapha* heaviness & big size never manifest.

श्लीपद लक्षण (Ślīpada Lakṣaṇa)

यः सज्वरो वङ्क्षणजो भृशार्तिः शोथो नृणां पादगतः क्रमेण ।
तच्छ्लीपदं स्यात् करकर्णनेत्रशिशनौष्ठनासास्वपि केचिदाहुः ॥

M.Ni. 39/1

Disease *Ślīpada* is characterised by fever; swelling starts from groins and gradually extend to feet associated with pain. Some say it may manifest in other places like hands, ears, eyes, penis, lips and nose.

वातज श्लीपद (Vātaja Ślīpada)

तत्र, वातजं खरं कृष्णं परुषमनिमित्तानिलरुजं परिस्फुटति
च बहुशः । S.S.Ni 12/14

वातजं कृष्णरूक्षं च स्फुटितं तीव्रवेदनम् ।
अनिमित्तरुजं तस्य बहुशो ज्वर एव च ॥

M.Ni 39/2

Vātaja Ślīpada Signs & Symptoms are as follows : Affected area becomes rough, blackish, coarse, has various kinds of *Vātaja* pain like piercing, pricking etc, severe pain appearing without any apparent cause, excessive crackness & fever.

पित्तज श्लीपद (Pittaja Ślīpada)

पित्तजं तु पीताव-भासमीषन्मृदुज्वरदाहप्रायं च ।

S.S.Ni 12/14

Pittaja Ślīpada Signs & Symptoms are as follows : Affected area becomes yellowish, slightly soft, associated with fever & burning sensation.

कफज श्लीपद (Kaphaja Ślīpada)

श्लेष्मजं तु श्वेतं सिग्धावभासं मन्दवेदनं भारिकं महाग्रन्थिकं
कण्टकैरुपचितं च ॥ S.S.Ni 12/14

Signs & Symptoms of Kaphaja Ślīpada are as follows : Affected area becomes whitish,

appears unctuous, has mild pain, heaviness, bigger nodules and studded with thorny projections.

साध्यासाध्यता (Sādhyāsādhyaṭā)

Prognosis of Ślīpada

तत्र संवत्सरातीतमतिमहद्वल्मीकजातं प्रसृतमिति वर्जनीयानि ॥

S.S.Ni 12/12

If *Ślīpada* persist for more than one year, growing bigger & bigger resembling like ant hill & exuding fluids should be rejected.

यच्छ्लेष्मलाहारविहारजातं पुंसः प्रकृत्याऽपि कफात्मकस्य ।
सास्त्रावमत्युन्नतसर्वलिङ्गं सकण्डुरं श्लेष्मयुतं विवर्ज्यम् ॥

M.Ni. 39/7

Ślīpada manifest due to consumption of foods & habits which aggravates *Kapha*, *Kaphaja Prakṛti*, exuding bigger in size, severe itching associated with all the symptoms should be rejected for treatment.

उपशय/पथ्य (Upaśaya/Pathya)

Process of Treatment: *Laṅgana*, external application, *Svedana*, *Virecana*, blood-letting and such other therapies as are hot and alleviator of *Kapha* should be used for the treatment of *Ślīpada* (filariasis). (B.R. 45/1)

The following measures and edibles are considered to be beneficial for the patient suffering from the disorder of *Ślīpada*: *Vamana*, *Laṅgana* (fasting), *Rakta Mokṣana* (blood letting), *Svedana*, *Virecana*, *Lepa*, old *Ṣaṣṭhi* variety of rice, *Yava* (barley), *Kulattha*, garlic, *Paṭola*, brinjal, fruits of *Śobhāñjana*, *Kāravellaka*, *Punarnavā*, radish, *Pūtikā* leaves, oil of *Eraṇḍa*, cow urine, *Kaṭu* and the like foods which promote digestive power. (B.R. 45/57-58)

अनुपशय/अपथ्य (Anupaśaya/Apathya)

The following measures and edibles are considered to be harmful for the *Ślīpada* patients: *Piṣṭāṇna*, *Dugdhavikṛti* (milk products), *Guda* (jaggery), meat of animals and birds living in marshy land, sweet juices, river water evolving

from the mountains of Mahendra, Sahya and Vindhya, oily products, heavy to digest and *Abhisyandi* products. (B.R. 45/59-60)

Filariasis/ lymphatic filariasis

Introduction

1. Filarial worms are tissue-dwelling nematodes.
2. The larval stages are inoculated by biting mosquitoes or flies, each specific to a particular filarial species.
3. The larvae develop into adult worms (2-50 cm long) which, after mating, produce millions or microfilariae (170-320 mm long) that migrate in blood or skin.
4. The life cycle is completed when the vector takes up microfilariae while feeding on humans.
5. In the insect, ingested microfilariae develop into infective larvae for inoculation in humans, normally the only host.
6. Disease is due to the host's immune response to the worms (both adult and microfilariae), particularly dying worms, and its pattern and severity vary with the site and stage of each species.
7. The worms are long-lived; microfilariae survive 2-3 years and adult worms 10-15 years.
8. The infections are chronic and worst in individuals constantly exposed to reinfection.

Causes

Infection with the filarial worms *Wuchereria bancrofti* and *Brugia malayi* is associated with clinical outcomes ranging from subclinical infection to hydrocele and elephantiasis

Mode of transmission

1. *W. bancrofti* is transmitted by night-biting culex or anopheline mosquitoes in most areas.
2. The adult worm, 4-10 cm in length, live in the lymphatics, and the females produce microfilariae which circulate in large numbers in the peripheral blood, usually at night.
3. The infection is widespread in tropical Africa, the North African coast, coastal areas of Asia, Indonesia and Northern Australia, the South

Pacific islands, the West Indies and also in North and South America.

Pathogenicity of filarial infections depending on site and stage of worms

Worm species	Adult worm	Microfilariae
<i>Wuchereria bancrofti</i> and	Lymphatic vessels-	Blood+ Pulmonary
<i>Brugia malayi</i>	Subcutaneous+	capillaries+++ Blood+
<i>Onchocerca volvulus</i>	Subcutaneous+	Skin+++ Eye+++
<i>Mansonella perstans</i>	Retroperitoneal-	Blood+
<i>Mansonella streptocerca</i>	Skin+	Skin++

(+++ severe; ++ moderate; + mild; - rarely pathogenic)

Pathology

Toxins, released by the adult worm cause lymphangiectasia



The dilatation of the lymphatic vessels leads to lymphatic dysfunction



Leading to lymphatic filariasis, lymphoedema and hydrocele



Death of the adult worm results in acute filarial lymphangitis.



The filariae are symbiotically infected with rickettsia-like bacteria (*Wolbachia* spp.)



Release of lipopolysaccharide from these bacteria contributes to inflammation.



Lymphatic obstruction persists after death of the adult worm.



Secondary bacterial infections cause tissue destruction.



The host response to microfilariae is central to the pathogenesis of tropical pulmonary eosinophilia.

Clinical features

1. Acute filarial lymphangitis presents with fever, pain, tenderness and erythema along the course of inflamed lymphatic vessels.
2. Inflammation of the spermatic cord, epididymis and testis is common.
3. The whole episode lasts a few days but may recur several times a year.
4. Temporary oedema becomes more persistent and regional lymph nodes enlarge.
5. Progressive enlargement, coarsening, corrugation, fissuring and bacterial infection of the skin and subcutaneous tissue develop gradually, causing irreversible 'elephantiasis'.
6. The scrotum may reach an enormous size.
7. Chyluria and chylous effusions are milky and opalescent; on standing, fat globules rise to the top.
8. The acute lymphatic manifestations of filariasis must be differentiated from thrombophlebitis and infection.
9. The oedema and lymphatic obstructive changes must be distinguished from congestive cardiac failure, malignancy, trauma and idiopathic abnormalities of the lymphatic system.
10. *Non-filarial elephantiasis* : Silicates absorbed from volcanic soil can also cause non-filarial elephantiasis.

Diagnosis

In the earliest stages of lymphangitis the diagnosis is made on clinical grounds, supported by eosinophilia and sometimes by positive filarial serology.

Investigations

1. *Full Blood Count (FBC)* : Filarial infections cause the highest eosinophil counts of all helminthic infections.
2. *Blood for Microfilariae* : Microfilariae can be found in the peripheral blood at night
3. Microfilariae are seen moving in a wet blood film or are detected by microfiltration of a sample of lysed blood.
4. *Hydrocele fluid examination* : They are

usually present in hydrocele fluid, which may occasionally yield an adult filaria.

5. By the time elephantiasis develops, microfilariae become difficult to find.
6. *Radiography* : Calcified filariae may sometimes be demonstrable by radiography.
7. *Scrotal ultrasound* : Movement of adult worms can be seen on scrotal ultrasound.
8. *PCR-based tests* : PCR-based tests for detection of *W. bancrofti* and *B. malayi* DNA from blood have been developed.
9. *Indirect fluorescence and ELISA* : Indirect fluorescence and ELISA detect antibodies in over 95% of active cases and 70% of established elephantiasis.
10. *Immunochromatographic card tests* : Highly sensitive and specific immunochromatographic card tests for detection of filarial infection are now commercially available; finger prick blood taken at any time of the day can be used for these.

Tropical pulmonary eosinophilia

1. Tropical pulmonary eosinophilia is a complication of Filarial infections seen mainly in India
2. It is due to microfilaria trapped in the pulmonary capillaries and destroyed by allergic inflammation.

Clinical features

1. It is characterised by paroxysmal cough, wheeze and fever.
2. If untreated, this progresses to debilitating chronic interstitial lung disease.

Investigations

1. In tropical pulmonary eosinophilia, serology is strongly positive and IgE levels are massively elevated but circulating microfilariae are not found.
2. The chest X-ray shows military changes or mottled opacities.
3. Pulmonary function tests show a restrictive picture

MCQs

1. Which layer of skin is the *Adhishtana* of *Shleepada*
 - a. *Tamra*
 - b. *Vedini*
 - c. *Rohini*
 - d. *Mamsadhara*
2. *Dushyas* involved in *Shleepada*
 - a. *Rakta Mamsa*
 - b. *Rasa Rakta Mamsa*

- c. *Rakta Mamsa Medas*
- d. *Mamsa Medas*
3. Types of *Shleepada*
 - a. 3
 - b. 4
 - c. 5
 - d. 7
4. *Kandakairupachitam* is the feature of
 - a. *Vataja Sleepada*
 - b. *Pittaja Sleepada*
 - c. *Kaphaja Sleepada*
 - d. Both a & b

Answer

1.	c	2.	a	3.	a
4.	c				